

Put Gravity to Work

Student Guide and Data Dictionary

The Challenge

You'll be working with real WNBA data from AWS Inside the Game's new **Gravity** stat. Pick one of two routes and build something that shows what Gravity reveals about the game.

The Fan

Build something that helps the fans see the game differently. Your audience could be a diehard, someone watching their first game, or anyone in between. Show them why **Gravity** matters and what it reveals that a box score can't.

The Front Office

Think like a team data scientist. Use **Gravity** to break down a player or matchup for your front office or coaching staff: visualize how she bends the defense, where her impact shows up, and how you'd game plan for her (or against her).

Show us: one clear insight, how you found it, and who it's for.

How judging works

Mentors and judges will check in on every team during the jam. At the end, the top 2-3 projects in each route will present to the judges for 5 minutes. Build with that in mind:

- One clear insight. What did you find?
- How you found it. Which data, and what did you do with it?
- Who it's for, and why they should care.
- At least one visualization that makes your point on its own.

What Is Gravity?

Gravity measures how much a player pulls defenders out of their normal assignments. In other words, how much attention she draws compared to what the spacing on the floor would predict. It tells us who forces defenses to react, even before they touch the ball.

How it Works

- The WNBA's optical tracking system follows 20 points on every player's body, 25 times per second (each snapshot is called a frame).
- For every frame, an AWS machine learning model estimates how much defensive pressure a player **should** get given where she is and where the ball is (Expected Defensive Pressure).
- It compares that to the pressure she **actually** gets (Defensive Pressure). The gap between the two is her Gravity.



OCTOBER 8, 2026

Interpreting Scores

Gravity score	What it means
Below 0	The defense gives her less attention than expected.
0	League average.
Above 0	She's got pull. Defenders are reacting to her.
Above 80 (off-ball)	Defenses are worried about her before she even catches the ball.
Above 90	She's bending the floor. The defense can't take its eyes off her.

Scores run from -100 to 100. Keep in mind those thresholds describe single moments. Averaged over a whole game or season, numbers get pulled toward 0, so a season average of +8 is elite.

Four Situations

Gravity is also split into four situations, so you can see where a player draws attention:

Situation	Description
On-ball perimeter	She has the ball, away from the basket.
Off-ball perimeter	She doesn't have the ball, away from the basket (spotting up, running off screens).
On-ball interior	She has the ball, close to the basket.
Off-ball interior	She doesn't have the ball, close to the basket (posting up, rolling, crashing).

More Information: <https://www.wnba.com/news/intro-to-gravity-stat-wnba-2026>

WNBA MINI DATA JAM powered by AWS

OCTOBER 8, 2026

Data Dictionary

You get three Gravity files that go from zoomed out to zoomed in, plus a player lookup file for names:

File	Rows	One row is...
wnba_gravity_season_leaderboards	69	One player's 2026 regular season (qualified players only), with names, team, and basic stats
wnba_gravity_by_game	5,521	One player in one game (269 games, 239 players, all 2026 regular season)
wnba_gravity_sample	108,560	One player in one frame of a single game: Indiana Fever at Minnesota Lynx, Aug. 2, 2026
wnba_player_lookup	239	One player: name, team, school, country, and draft info for every player in the other files
wnba_team_lookup	15	One team: ID, abbreviation, city, and name

How the Files Connect

Key	Links
playerId	Same player ID in every file. Join to the player lookup to get names for anyone in the game or sample files.
Game_ID / gameId	Game ID in the game file (Game_ID) and the sample file (gameId). The sample game is 1022600218.
teamId / Team_ID	Team ID in the game file (teamId) and leaderboard (Team_ID). Use the team table on the last page to get names.

File 1: wnba_gravity_season_leaderboards

Season totals for the 69 players who qualified (every player here played at least 21 games).

Column	Type	What it means
playerId	number	Unique player ID
First_Name, Last_name	text	Player name
Jersey_Num	number	Jersey number
Team_ID	number	Unique team ID
Team_City, Team_Name, Team_Abbreviation	text	Team info (for example Toronto, Tempo, TOR)
Frames	number	Total frames tracked for this player all season. More frames means more time on the floor.



WNBA MINI DATA JAM powered by AWS

OCTOBER 8, 2026

Column	Type	What it means
Gravity	decimal	Season Gravity: the average of her game-by-game Gravity scores. Every game counts equally.
GravityScoreAvg	decimal	Season Gravity averaged across every frame instead. Games where she played more count more. Very close to Gravity for most players.
onballperimeter_Frames	number	Frames she spent with the ball on the perimeter
onballperimeter_Gravity	decimal	Gravity with the ball on the perimeter (game average)
onballperimeter_GravityScoreAvg	decimal	Gravity with the ball on the perimeter (frame average)
offBallPerimeter_*, onBallInterior_*, offBallInterior_*		Same three columns (Frames, Gravity, GravityScoreAvg) for the other three situations
GP	number	Games played
MPG, PPG, RPG, APG	decimal	Minutes, points, rebounds, and assists per game

File 2: wnba_gravity_by_game

One row per player per game. No player names here, so join to the player lookup on playerId.

Column	Type	What it means
Game_ID	number	Unique game ID
teamId	number	The player's team in this game
playerId	number	Unique player ID
frames	number	Frames tracked for this player in this game
averageGravity	decimal	Her Gravity for the game, averaged across all her frames
onBallPerimeter_Frames	number	Frames with the ball on the perimeter
onBallPerimeter_averageGravity	decimal	Gravity with the ball on the perimeter
offBallPerimeter_Frames / _averageGravity		Same, off the ball on the perimeter
onBallInterior_Frames / _averageGravity		Same, with the ball in the interior
offBallInterior_Frames / _averageGravity		Same, off the ball in the interior
league_id	number	League code. Always 10 (WNBA).
Season_ID	number	Season code. Always 22026 (2026 regular season).



File 3: wnba_gravity_sample

Frame-by-frame tracking for one full game: Indiana Fever at Minnesota Lynx on Aug. 2, 2026. Minnesota won 108-100. Each frame has 5 rows, one for each player on offense. Frames are 0.08 seconds apart (every other frame from the 25-per-second feed).

Column	Type	What it means
(first column, no name)	number	Row number. You can ignore it.
frameId	number	Frame ID. All 5 rows in the same moment share it.
gameId	number	Game ID (1022600218)
gameDate	date	Game date (2026-08-02)
season	number	Season (26 = 2026)
period	number	Quarter (1 to 4)
gameClockTime	text	Time left in the quarter. Watch out: shown as minutes:seconds (9:57) above one minute, and as seconds (49.72) under one minute.
shotClockTime	decimal	Seconds left on the shot clock
timeUTC, unixTime	text / number	Real-world timestamp of the frame
ballCentroid	text	Ball location as [x y z] in feet. z is height off the floor.
playerId	number	Unique player ID
playerCentroid	text	Player location on the court as [x y] in feet
ballHandler	True/False	True if this player has the ball. Exactly one per frame.
hasGravity	True/False	False if Gravity wasn't calculated for this frame
playerGravity	decimal	Her Gravity in this exact moment (-100 to 100). Blank when hasGravity is False.

About the court coordinates

- Locations are in feet, with (0, 0) at center court.
- x runs the length of the court, baseline to baseline (-47 to 47). The hoops sit at about $x = -41.75$ and $x = 41.75$.
- y runs the width of the court, sideline to sideline (-25 to 25).
- Ball z is the ball's height off the floor in feet. The rim is 10 feet up.
- Teams switch baskets at halftime. In the sample game, Minnesota attacks the +x hoop in quarters 1 and 2 and the -x hoop in quarters 3 and 4 (Indiana is the opposite). To put every possession on one half court, flip the signs of x and y for the -x half.
- Coordinates are stored as text like [-3.63 17.04 1.79], so split them into separate x, y, and z columns before plotting.

File 4: wnba_player_lookup

One row for each of the 239 players who appear in the Gravity files.

Column	Type	What it means
playerId	number	Unique player ID
First_Name, Last_Name, Full_Name	text	Player name
School	text	College or last school/club before the WNBA (blank for some international players)
Country	text	Home country
Draft_Year, Draft_Round, Draft_Number	number	WNBA draft info. Blank means she went undrafted.
Rookie_Year	number	First WNBA season
Team_ID, Team_Abbreviation	number / text	Her most recent 2026 team in the data
All_Teams_2026	text	Every 2026 team she played for in the data, in order (for example PHX / LAS if she was traded)
Games_In_Data	number	Number of games she appears in the game file
In_Leaderboard	True/False	True if she's one of the 69 qualified players in the leaderboard
In_Sample_Game	True/False	True if she appears in the frame-by-frame sample game

File 5: wnba_team_lookup

One row for each of the 15 WNBA teams.

Column	Type	What it means
Team_ID	number	Unique team ID. Matches teamId in the game file and Team_ID everywhere else.
Team_Abbreviation	text	Three-letter team code (for example MIN)
Team_City	text	Team city or region (for example Minnesota)
Team_Name	text	Team nickname (for example Lynx)
Team_Full_Name	text	City and nickname together (for example Minnesota Lynx)

Things to Watch Out For

- **Small samples swing wildly.** A player with only a few frames in a game can show up at exactly -100 or 100. Set a minimum (for example, only games with 500+ frames) before ranking anyone.
- **Same for the four situations.** A center might only have a handful of on-ball perimeter frames. Check the _Frames column before trusting a split.
- **Frames aren't minutes.** Frames are only counted when Gravity was calculated, so don't convert them to playing time. Use MPG from the leaderboard for that.

Ideas to Get You Started

You don't have to use any of these. They're here if you're stuck.

Fan Route

- A "Gravity card" for a favorite player that a casual fan would understand in five seconds.
- Who has more pull than her points per game would suggest? Find the hidden stars.
- Animate a few possessions from the sample game and show Gravity changing as players move.
- Take an Educational route. Explain Gravity to someone who has never watched basketball.
- Team identity: which teams lean on off-ball movement vs. players creating with the ball?

Team Route

- Scouting report: where does a player draw the most attention, and how would you defend her?
- Does a player's Gravity change depending on the opponent? Who shuts her gravitational pull down?
- Lineup spacing: which teammates have high off-ball Gravity, and does that open things up for the ball handler?
- Using the sample game: what happens to Gravity late in the shot clock or late in the game?
- Find a role player whose off-ball Gravity makes her more valuable than her box score shows.

New to Basketball Data and Analysis?

- **Start with the leaderboards.** It has names, teams, and basic box score stats in one place, and it opens easily in a spreadsheet.
- **Want trends or matchups?** Use the game file to see how a player's Gravity changes game to game or against certain teams.
- **Want to get on the court?** The sample file has player and ball locations frame by frame, so you can plot where players were standing and see Gravity move in real time.